MILITARY STANDARD

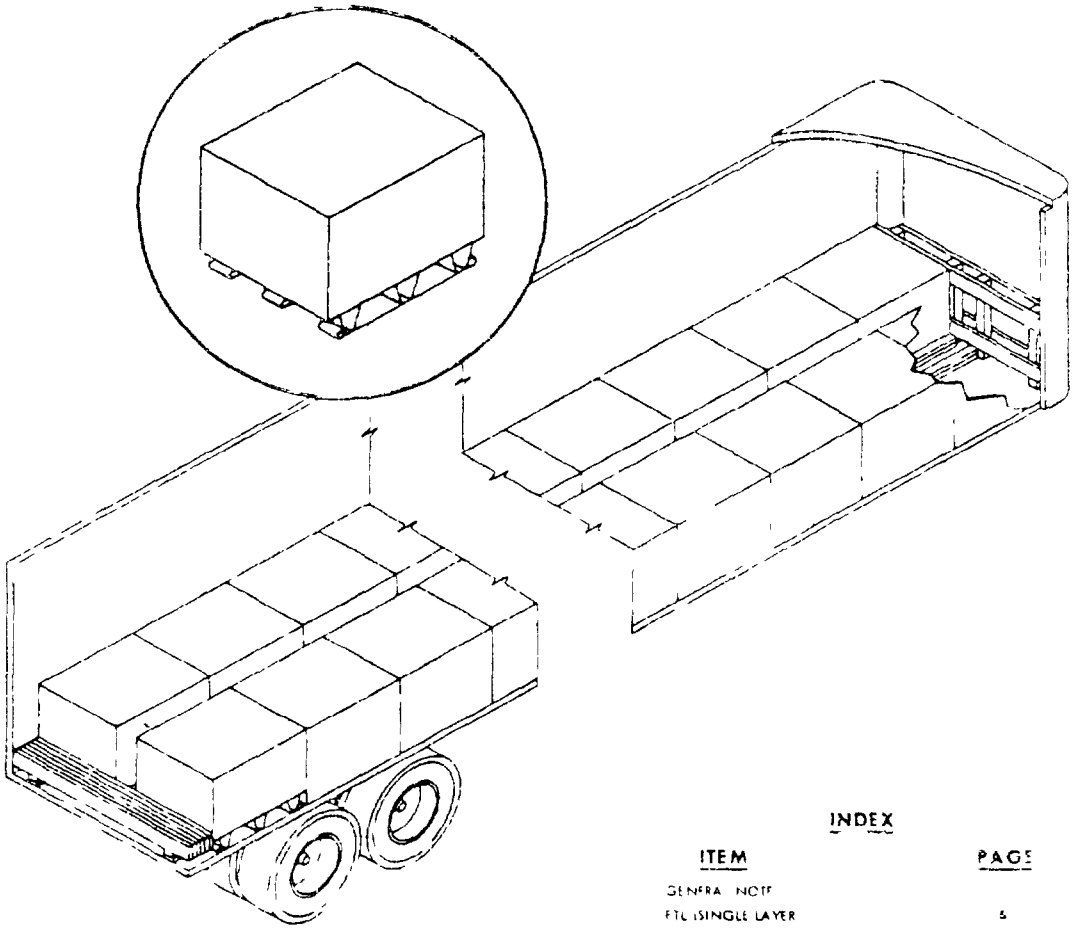
TRUCKLOADING OF HAZARDOUS MATERIALS

PALLETIZED UNIT LOADS
DOUBLE ROW PATTERN

MIL-STD-1320 2
(NAV.)

29 JANUARY 1975

SUPERSEDING
WE N 2A
22 AUGUST 967



INDEX

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GENERA NOTE	
FTL (SINGLE LAYER	5
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DETAIL	6
LTL	
UNITIZ NG DETA	

NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING CONTAINER LOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL HDBK-236

FSC 8140

AUTHORIZED AND RELEASED
FOR HIGHWAY SHIPMENTS
ONLY

SIGNATURE

ORIGINATOR

NAVAL WEAPONS HANDLING LABORATORY
NAVAL WEAPONS STATION EARLE NEW JERSEY

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GENERAL NOTES

THE INDEX TO STANDARDS FOR PALLETIZING TRUCKLOADING RAILCAR LOADING ND CONTAINER LOADING MIL-HDBK 236 (NAVY LISTS WEAPON COMPONENTS AND INDICATES THE CORRECT DOCUMENT TO BE USED IN TRUCKLOADING OF EACH ITEM INDEXED MIL-HDBK-236 INDEXES THE UNIT LOADS THAT ARE AUTHORIZED TO BE TRUCKLOADED IN ACCORDANCE WITH THIS DOCUMENT

4 HIS DOCUMENT PROVIDES DETAILED TRUCKLOADING INSTRUCTIONS APPLICABLE TO PALLETIZED UNIT LOADS WHEN THE WIDTH OF THE UNIT LOAD PERMITS LOADING IN THE PATTERN PRESCRIBED

PROCEDURES AND PRACTICES CONTAINED HEREIN ARE INTENDED FOR VAN-TYPE TRAILERS OF ALL LENGTHS THE TRAILERS MAY HAVE WOOD FLOORS METAL FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS

THE BLOCKING METHOD SHOWN ON PAGES 5 & ARE FOR WOOD FLOOR TRAILERS WITH UNIT LOADS OF DIMENSIONS THAT PERM THE PROCEDURES SHOWN WHEN THE TRAILER HAS METAL FLOORS METAL FLOORS WITH NAILING STRIPS OR THE CHARACTERISTIC OF UNIT LOAD MAKES BLOCKING AND BRACING SHOWN IMPRACTICABLE TO USE CHOOSE AN APPROPRIATE ALTERNATE METHOD AS EXPLAINED IN THE SPECIFIC INSTRUCTIONS FOR THE LOAD

A FULL TRUCK LOAD (FTL) CONSISTS OF AS MANY UNIT LOADS AS CAN BE ARRANGED IN THE TRAILER CONSIDERING ITS CUBIC PERMISSIBLE GROSS WEIGHT AND AXLE LOAD LIMITATIONS IF THESE LIMITATIONS PERMIT UNIT LOADS MAY BE DOUBLE OR TRIPPLE LAYER IN ACCORDANCE WITH THE PRINCIPLES SET FORTH ON PAGES 4 AND 5

6 A LESS-THAN-TRUCK LOAD (LTL) SHOULD BE ARRANGED CONSIDERING THE WEIGHT DISTRIBUTION ON THE TRUCK A TYPICAL LTL IS SHOWN ON PAGE 6

AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA CLOSE AND SEAL TRAILER DOORS AND ATTACH APPROPRIATE PLACARD (IF REQUIRED) TO BOTH SIDES FRONT AND BACK OF TRAILER

8 APPLICABLE MATERIAL SPECIFICATIONS DUNNAGE LUMBER MM-L-751 NAILS FF-N-105 TYPE II STYLE 10 COMMON BRIGHT STRAPPING QQ-S-781 TYPE I CLASS A

7 FOR GENERAL TRUCK LOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)

PROCEDURE (SINGLE LAYER)

THE LOAD SHOWN ON PAGE 3 IS INTENDED TO ILLUSTRATE TYPICAL BLOCKING AND BRACING PROCEDURES FOR A SINGLE LAYER LOAD IN A WOOD FLOOR TRAILER FOR TRAILERS WITH METAL FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS USE THE APPROPRIATE AUTHORIZED ALTERNATE BLOCKING METHOD SEE NOTE 3 BELOW

WARNING

DO NOT NAIL BLOCKING OR BRACING INTO METAL FLOOR

WHEN SPACE BETWEEN PALLETS DOES NOT PERMIT SLEEPERS AGAINST EACH PALLET AS SHOWN ON PAGE 3 OR INSTALLATION IS DIFFICULT A DOUBLED SLEEPERS OF TWO INCH NOMINAL THICKNESS AND APPROPRIATE WIDTH(S) MAY BE USED

NOTE

WHEN THE SPACE BETWEEN THE UNIT LOADS IS LESS THAN 4 INCHES BLOCKING TO PREVENT LATERAL MOTION MAY BE OMITTED

5 WHEN THE TRAILER HAS METAL FLOORS OR WOOD NAILING STRIP THAT WILL NOT ACCOMODATE SLEEPER USE SWAY BRACES TO PREVENT LATERAL MOTION AS SHOWN FOR THE LOWER UNIT LOADS IN DETAIL G PAGE 6

4 WHEN ALL LOAD ARE IN PLACE INSTALL REAR BLOCKING TYPE REQUIRED IS DEPENDENT UPON DISTANCE BETWEEN TRAILER DOOR WHEN CLOSED AND REAR OF LOADING DETERMINE THIS DISTANCE AND INSTALL BLOCKING AS INDICATED BELOW

DISTANCE LESS THAN 12 INCHES

INSTALL BLOCKING DETAIL B AS SHOWN ON PAGE 7

CAUTION

REAR BLOCKING MUST BEAR AGAINST TRAILER DOOR WHEN DOOR IS IN CLOSED POSITION DO NOT USE TRAILERS WITH ROLL UP DOORS

DISTANCE 12 TO 36 INCHES

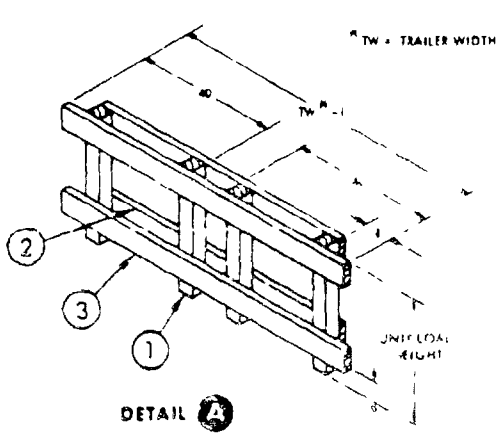
INSTALL BLOCKING DETAIL C AS SHOWN ON PAGE 7

CAUTION

REAR BLOCKING MUST BEAR AGAINST TRAILER DOOR WHEN DOOR IS IN CLOSED POSITION DO NOT USE TRAILERS WITH ROLL UP DOORS

DISTANCE GREATER THAN 36 INCHES

INSTALL BLOCKING AS SHOWN IN TYPICAL LTL FACE

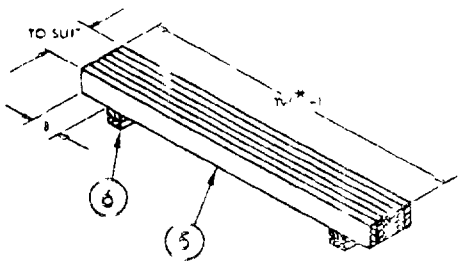


DETAIL A

FRONT BULKHEAD REQUIRED

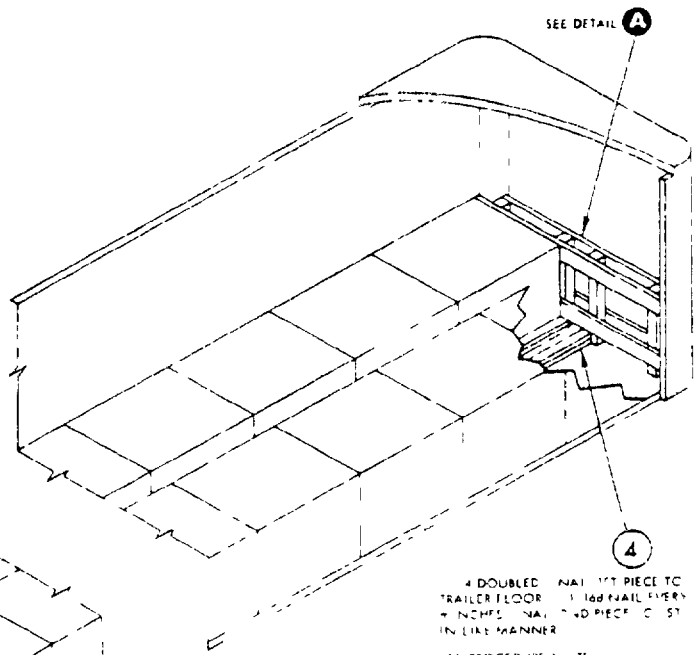
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* W = TRAILER WIDTH



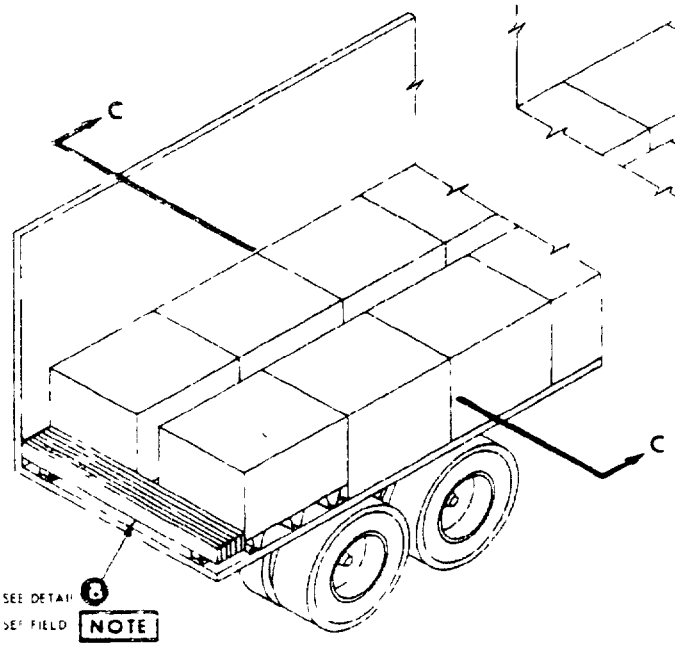
DETAIL B

REAR BLOCKING ASSEMBLY
1 REQUIRED



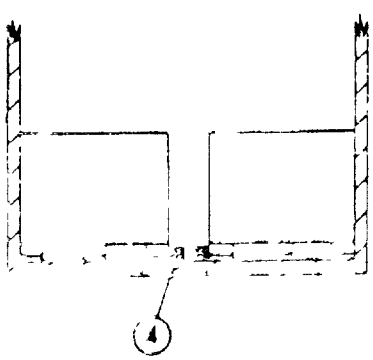
4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR 1 16d NAIL EVERY INCHES NAIL 2ND PIECE TO 1ST IN LIKE MANNER

SEE PROCEDURE NOTE



NOTE

THIS DRAWING ILLUSTRATES A LOAD WHEN THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOORS WHEN CLOSED 12 INCHES OR LESS. FOR SINGLE LAYER PROCEDURE REFER TO NOTE 4 FOR DETAILED INSTRUCTIONS FOR THIS AND OTHER SIMILAR



SECTION C-C

* W = TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
1	DOOR		4			
2	SOLID		4 REQD LAMINATE 1 PIECE			
3	CLEEPER		AS REQD	SEE FIELD NOTE		
4	APPROXIMATE				JOINT	2a
5	APPROXIMATE				JOINT	2a
6	VERTICAL					

ALL MATERIAL SHALL MEET MIL-STD-209A

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PROCEDURE (MULTI-LAYER)

THE LOAD SHOWN ON PAGE 5 IS INTENDED TO ILLUSTRATE TYPICAL BLOCKING AND BRACING PROCEDURES FOR A TWO LAYER LOAD IN A WOOD FLOOR TRAILER. FOR TRAILERS WITH METAL FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS USE THE APPROPRIATE AUTHORIZED ALTERNATE BLOCKING METHOD (SEE NOTE 5 BELOW).

WARNING

DO NOT NAIL BLOCKING OR BRACING INTO METAL FLOOR

2. TO PREVENT LONGITUDINAL MOVEMENT IN THE SECOND (AND THIRD) LAYER(S) WHERE THE LAYER CHANGES FROM TWO LAYERS TO ONE LAYER HIGH (OR THREE LAYERS TO TWO LAYERS HIGH) AND AT THE REAR OF THE LOAD WHEN TWO OR MORE LAYERS HIGH. THE UNIT LOADS SHALL BE UNITIZED AS SHOWN ON PAGE 6. THE STIFFENERS SHALL BE POSITIONED TOWARDS THE LOWER LAYER(S) OR WHEN AT THE REAR OF THE LOAD, TOWARDS THE REAR.
3. WHEN SPACE BETWEEN PALLETS DOES NOT PERMIT SLEEPERS AGAINST EACH PALLET AS SHOWN ON PAGE 5 OR INSTALLATION IS DIFFICULT, A DOUBLED SLEEPER(S) OF TWO INCH NOMINAL THICKNESS AND APPROPRIATE WIDTH(S) MAY BE USED.
4. USE SWAY BRACE, DETAIL H, PAGE 6 BETWEEN UNIT LOADS IN SECOND (AND THIRD LAYERS). WHEN SPACE BETWEEN UNIT LOADS DOES NOT PERMIT INSTALLATION OF SWAY BRACE, INSTALL FILLER ASSEMBLY, DETAIL J, PAGE 6. WHEN USING FILLER ASSEMBLY, SLEEPERS PIECE 4 ARE NOT REQUIRED.

NOTE

WHEN SPACE BETWEEN UNIT LOADS IS LESS THAN 4 INCHES
BLOCKING TO PREVENT LATERAL MOTION MAY BE OMITTED

5. WHEN TRAILER HAS FLOORS OF METAL OR METAL WITH WOOD NAILING STRIPS THAT WILL NOT ACCOMODATE SLEEPERS, USE SWAY BRACES TO PREVENT LATERAL MOTION AS SHOWN IN DETAIL G, PAGE 6 OR USE THE FILLER ASSEMBLY, DETAIL J, PAGE 6 AS APPROPRIATE.
6. INSTALL REAR BLOCKING, TYPE REQUIRED IS DEPENDENT UPON DISTANCE BETWEEN TRAILER DOOR WHEN CLOSED AND REAR LADING. DETERMINE THIS DISTANCE AND INSTALL BLOCKING AS INDICATED BELOW.

DISTANCE LESS THAN 12 INCHES

INSTALL BLOCKING DETAIL B AS SHOWN ON PAGE 5

CAUTION

REAR BLOCKING MUST BEAR AGAINST TRAILER DOOR WHEN DOOR IS IN CLOSED POSITION. DO NOT USE TRAILERS WITH ROLL UP DOORS.

DISTANCE 12 TO 36 INCHES

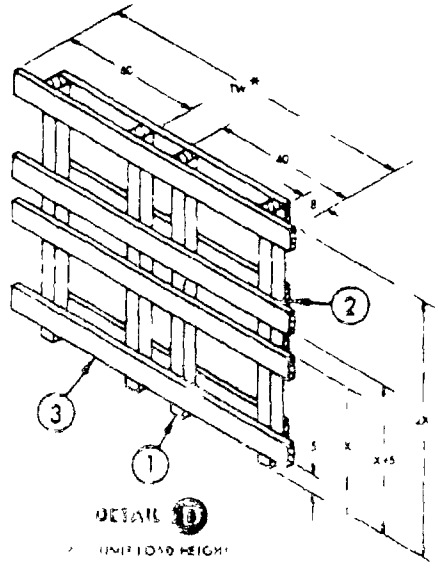
INSTALL BLOCKING DETAIL F AS SHOWN ON PAGE 5

CAUTION

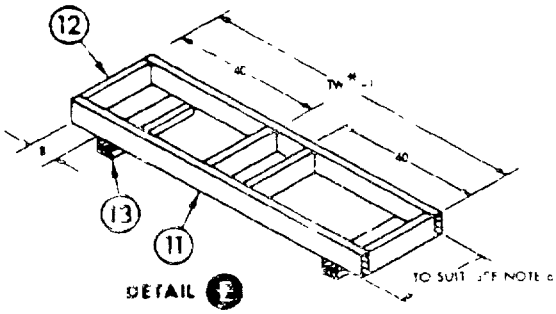
REAR BLOCKING MUST BEAR AGAINST TRAILER DOOR WHEN DOOR IS IN CLOSED POSITION. DO NOT USE TRAILERS WITH ROLL UP DOORS.

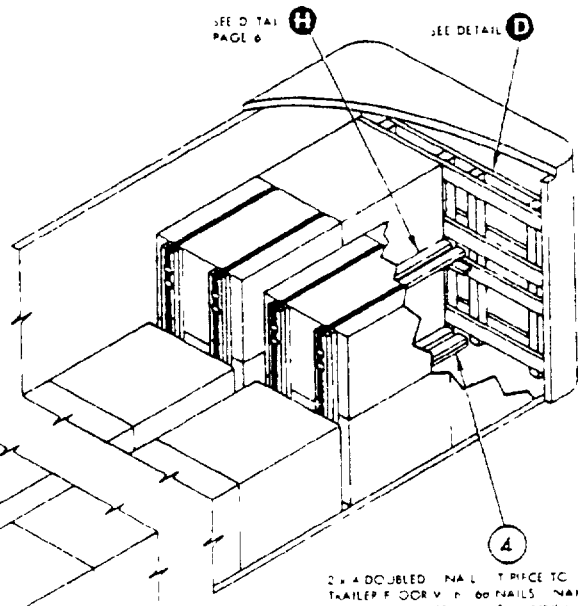
DISTANCE GREATER THAN 36 INCHES

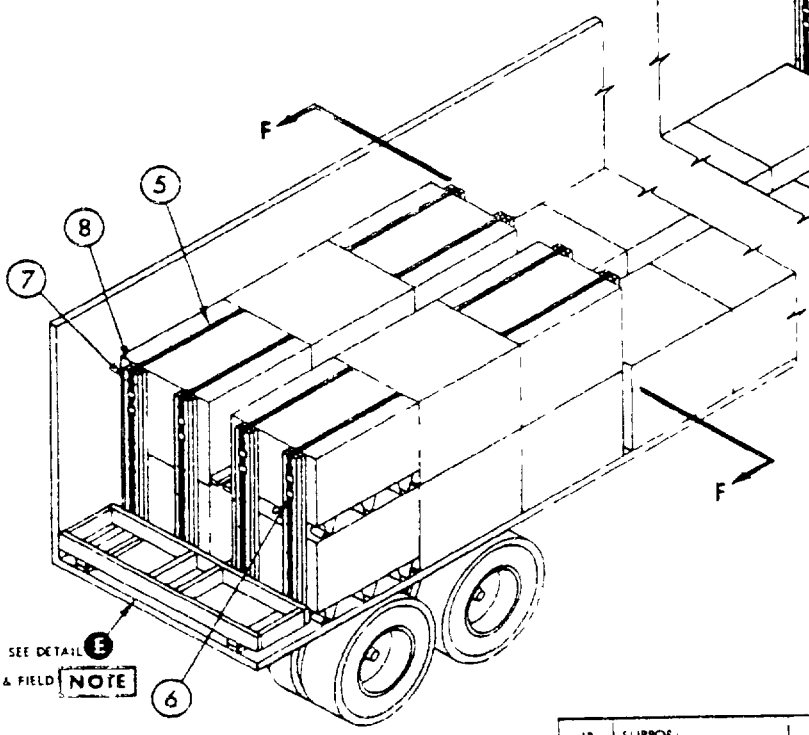
INSTALL BLOCKING AS SHOWN IN TYPICAL LIL PAGE



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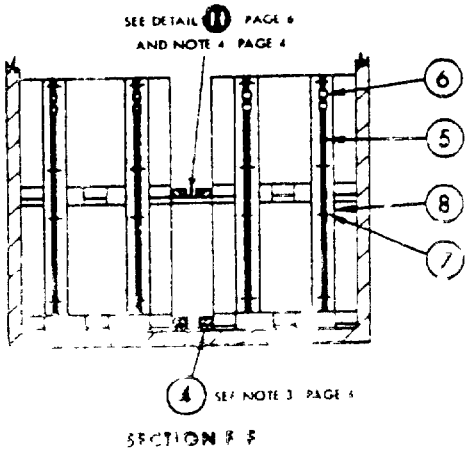






NOTE

THIS DRAWING ILLUSTRATES A LOAD WHEN THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOORS WHEN CLOSED IS 12 TO 36 INCHES SEE NOTE 6 PAGE 4 FOR DETAILED INSTRUCTIONS FOR THIS AND OTHER CONFIGURATIONS



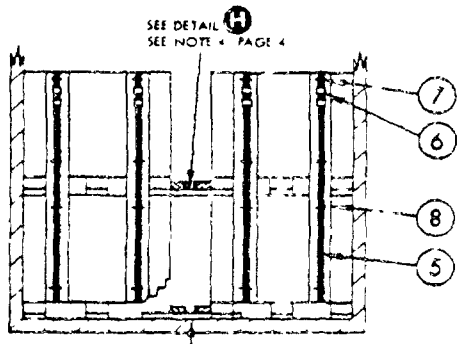
SECTION F F

*TW TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER NAILS	SIZE
13	SUPPORT	4 TO SUIT	6			
2	STRUT	4 TO SUIT	4	EE		
11	CROSSMEMBER	2 x 6 TW*			JOINT	60
10	STRINGER	4 x 4 INT LOAD LENGTH	6	9	JOIN	60
9	SUPPORT (DETAIL PAGE 6)	4 x 4 TO SUIT	10	SEE 10		
8	STIFFENER (DETAIL PAGE 6)	4 x 6 TO SUIT	24	LAMINATE		60
7	STAPLE	FOR 4 STRAP	48			
6	SEAL	FOR 4 TRA	4			
4	STRAP	FOR	2			
4	SLEEPER	2 4 TO SUIT	4 REQD	SEE FIELD NOTE		
3	AFT CROSSMEMBER	2 x 6 x TW*	4		TO AT	30
2	FWD CROSSMEMBER	2 6 TO SUIT	4		TO	4
	VERTICAL	4 4 TO SUIT	4	SEE NOTE	NUMBER	SIZE

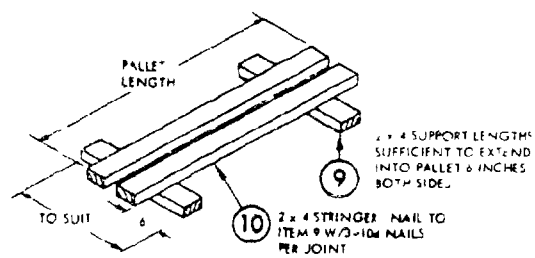
LIST OF MATERIALS AND NAILING DATA

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DETAIL G

SECTION OF A TWO LAYER HIGH LOAD SHOWING USE OF SWAY BRACES DETAIL H

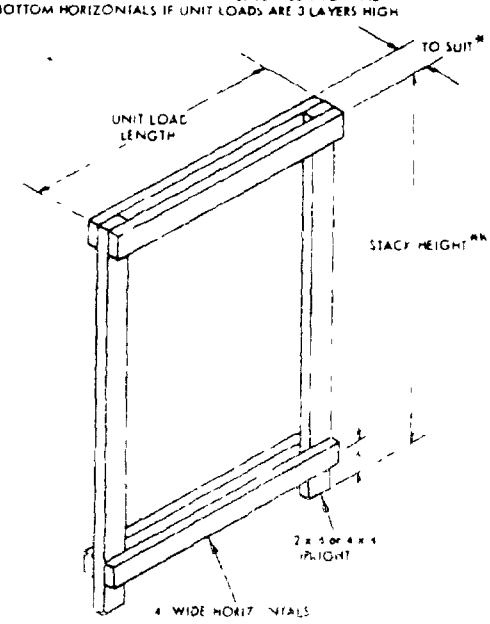


DETAIL H

SWAY BRACE ASSEMBLE IN PLACE

*HORIZONTALS MAY BE LAMINATED AND THICKNESS CHOSEN TO SUIT TO FILL VOID BETWEEN UNIT LOADS

**INSTALL THIRD SET OF HORIZONTALS BETWEEN TOP AND BOTTOM HORIZONTALS IF UNIT LOADS ARE 3 LAYERS HIGH



DETAIL J

FILLER ASSEMBLY (ALTERNATE FOR SWAY BRACE & EEPER)

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LTL ALL LENGTH TRAILERS

THE LTL SHOWN ON THIS PAGE IS INTENDED TO ILLUSTRATE TYPICAL BLOCKING AND BRACING METHOD FOR LTL ON WOOD FLOOR OR METAL WITH WOOD NAILING STRIP TRAILERS

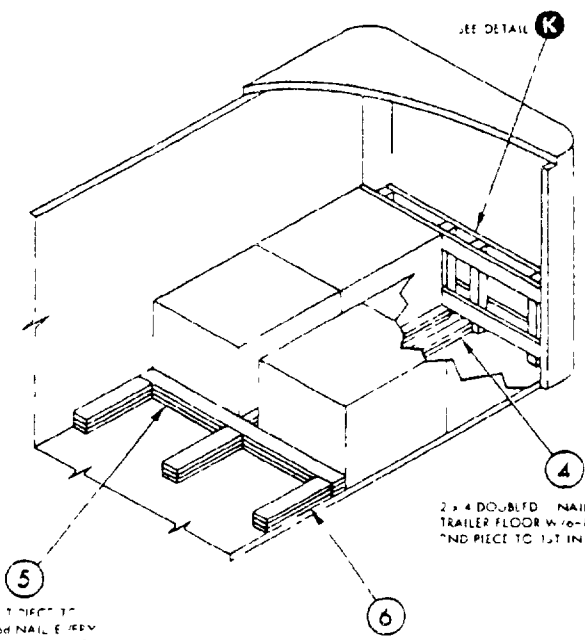
WARNING

DO NOT USE ALL METAL FLOOR TRAILERS FOR LTL SHIPMENTS AS THEY CAN NOT BE BLOCKED BY THIS METHOD

SWAY BRACE DETAIL C, FILLER ADJUSTABLE DETAIL J ARE AUTHORIZED ALTERNATES FOR SLEEPERS PIECE 2 AND IT MAY CONSIST OF MORE OR LESS UNIT LOAD AND ARE NOT NECESSARILY LIMITED TO THE ARRANGEMENT SHOWN

NOTE

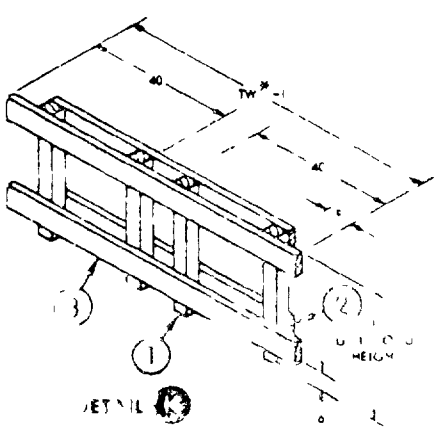
THIS DRAWING ILLUSTRATES A LOAD WHEN THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOORS GREATER THAN 36 INCHES THE TRAILER SHALL HAVE A WOOD FLOOR OR A METAL FLOOR WITH WOOD NAILING STRIPS



2 x 4 DOUBLED NAIL 5 PIECE TO TRAILER FLOOR W/6-16d NAILS NAIL 2ND PIECE TO 1ST IN LIKE MANNER

2 x 6 TRIPLED NAIL 3 PIECE TO TRAILER FLOOR W/ 16 6d NAILS EVERY 8 INCHES NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANN

2 x 6 TRIPLED NAIL 5TH PIECE TO TRAILER FLOOR W/ 6d NAILS NAIL 2ND PIECE TO 5TH PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER



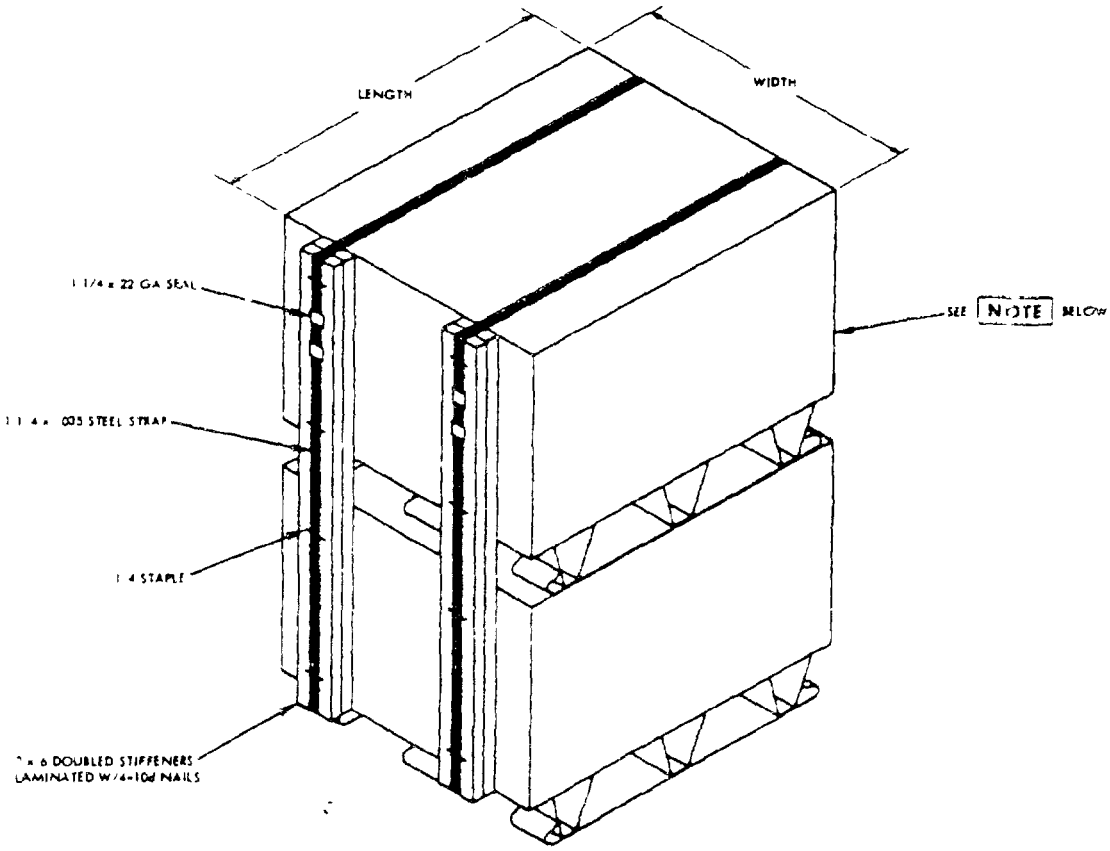
JOINT 2 PLY HEAD REQUIRED

*TW = TRAILER WIDTH

	BACK PIECE			SEE FIELD NOTE	
	CHOCK			SEE FIELD NOTE	
4	SLEEPER			SEE FIELD NOTE	
3	AFT CROSSMEMBER			JOINT	
2	FWD CROSSMEMBER			JOINT	
1	TRUCK		SEE		
PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL NO	NUMBER AILS

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UNITIZING DETAIL



1. WHEN REQUIRED BY THE FTL OR LTL REQUIREMENTS OF THIS MIL STD, THE STACKED UNIT LOADS SHALL BE UNITIZED AS SHOWN ABOVE. ADDITIONAL STIFFENERS SHALL EXTEND DOWN FROM THE TOP OF THE STACKED UNIT LOAD TO THE PALLET OF THE BOTTOM UNIT LOAD.
2. THE 1 1/4 INCH STEEL STRAPS POSITIONED AS SHOWN ENCIRCLE THE STACKED UNIT LOADS AND PASS UNDER THE DECK OF THE BOTTOM PALLET. THE STRAPS HOLD THE STIFFENERS IN PLACE.
3. [illegible] STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS. SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 INCH STAPLES.
4. WHEN [illegible] HIGH UNIT LOAD, FOLLOW THE PRINCIPLES OF THE TWO HIGH UNIT LOADS SHOWN ABOVE.

NOTE

WHEN LOADING A TRAILER WITH A SINGLE STACK OF UNIT LOADS, THE STIFFENERS SHALL BE AT BOTH ENDS OF THE UNITIZED LOADS.

PREPARING ACTIVITY
NAVY - YD
PROJECT NO. 4146-N2761

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

MIL-STD-1320-2 (Navy)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed)* *(Last, First, Middle Initial)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 58 WHICH MAY BE USED

S/N 0102-014-3801

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FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Naval Weapons Station Earle (803)
Naval Weapons Handling Laboratory
Colts Neck, New Jersey 07722

FOLD

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